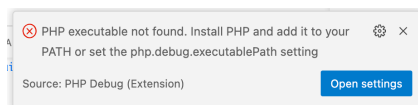
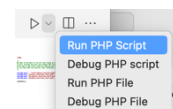


Refactoring Homework #5 For Testing

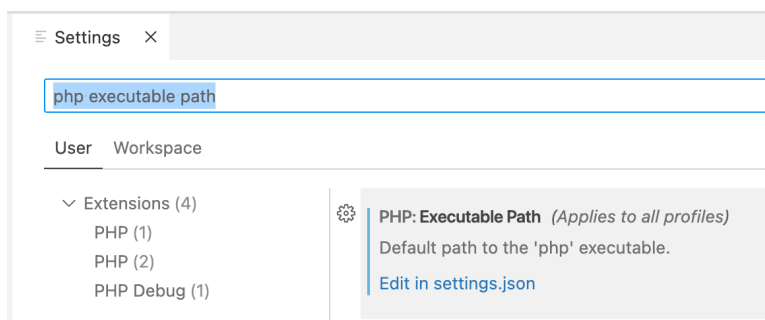
Homework #5 involved creating a tool for teachers using Brightspace to enter and persist the grades of their students. In Week 10, we learned about unit testing and how adding unit testing to your projects is a great way to test your code in an automated way. Rarely is software developed at companies without accompanying unit tests. There is, however, lots of code out there that does not have unit tests, including your answer to homework #5. This homework will involve adding unit tests to your solution to homework #5 and refactoring it as necessary to make the solution more testable. We already have gone over in class how the professor's answer to homework #5 was updated to unit test so use that as a template but do it against your code. You will find that once you start thinking about unit tests, you write code differently. You will use more functions, you will use more files, you will make functions independent (just the inputs and the output) so that they can more easily be unit tested. This is what I want to see. Do not add functionality to your solution or copy code from the answers, simply update your homework #5 code to make it more unit testable, with as many tests as can cover all the functionality. Now, if your homework #5 solution was incomplete or did not work, it is fine to fix it as part of this homework solution. But you will need to create a separate issue in GitHub with a separate branch and code review to handle fixing the solution before you do the work to add unit tests in a different branch and code review.

Getting VS Code to Run Unit Tests

Out of the box, VS Code does not know how to run PHP. If you select any PHP file and try to run it by clicking on the right facing arrow on the upper left, you will get an error message that the PHP executable is not found:



Click on Open Settings, or go into settings and type "php executable path" and click on the "Edit in settings.json" link:



Homework #6

You now need to edit this file to add the executable path for PHP, which comes from the XAMPP installation. The path will be different for Macintosh or Windows machines:

🍏 Macintosh Instructions

```
"php.executablePath": "/Applications/xampp/xamppfiles/bin/php"
```

🪟 Windows Instructions

```
"php.executablePath": "C:/xampp/php/php",
```

Save the file and you should now be able to run PHP in VS Code. If you run a regular PHP file that outputs HTML you will see in the output the HTML it produces. If you have a unit test you will see the output of the unit test in the debug console.

Instructions

1. If you have not already done so, version your homework #5 code in GitHub (make a revision that points to the code for the solution).
2. (Optional) Create a new issue to fix the solution if it did not work (see the feedback for homework #5). Focus on fixing the code, don't add functionality. Keep this code separate from code to add unit testing.
3. Create a new issue to add unit testing to the codebase and create a branch from it.
4. Add unit tests to test as much functionality as you can, refactoring the code base to make it easier to test. It's up to you to decide how much code to cover with unit tests. The goal is not to cover 100% but to do as much as possible to find any issues.
5. It's fine to fix any bugs your unit tests uncover.
6. Commit the changes to the branch, make a code review, and merge to main. Make a new revision of this codebase.
7. Submit a zip of the updated codebase to Brightspace and give me the GitHub link so I can see what was updated from one revision to the other.